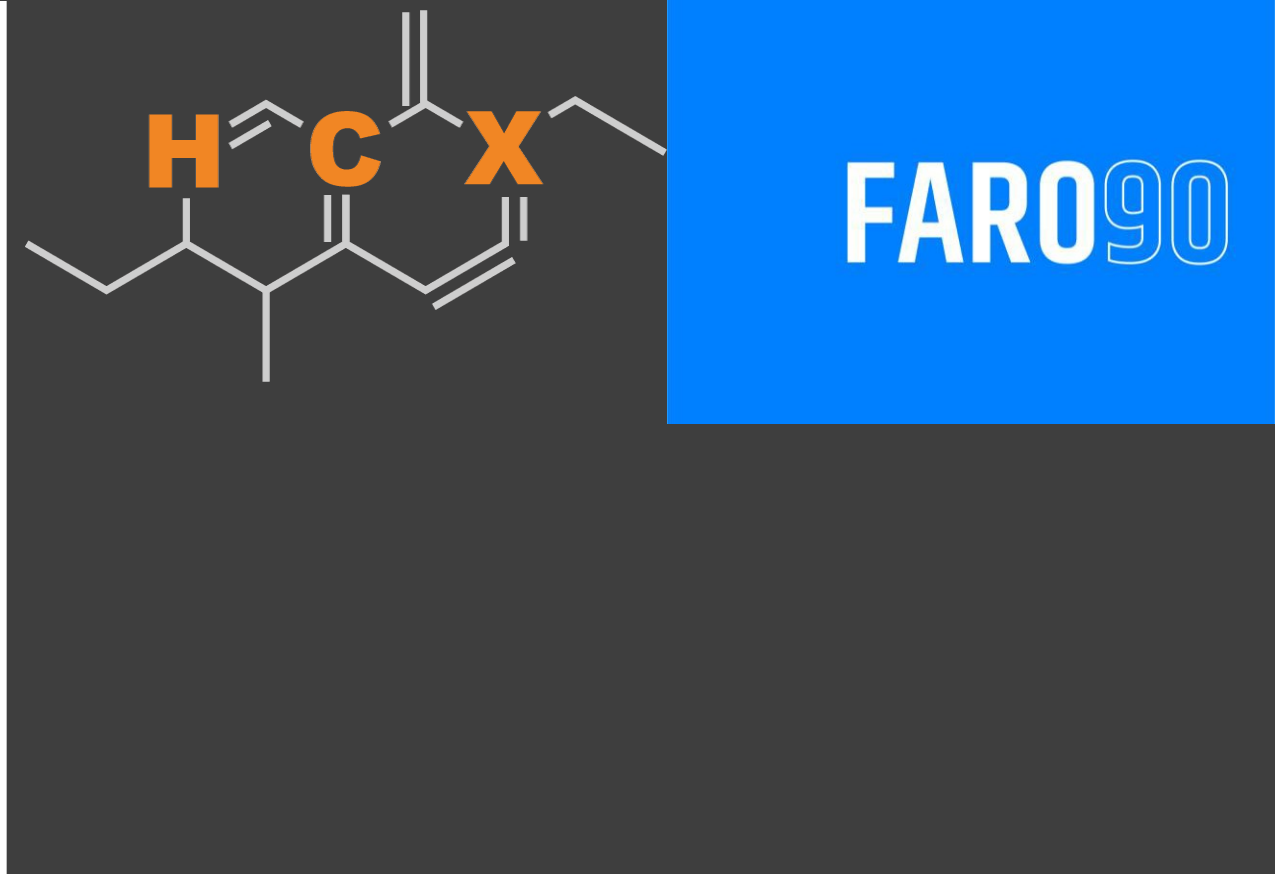




Ethanol Blending in Gasoline - Belgium

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Belgium

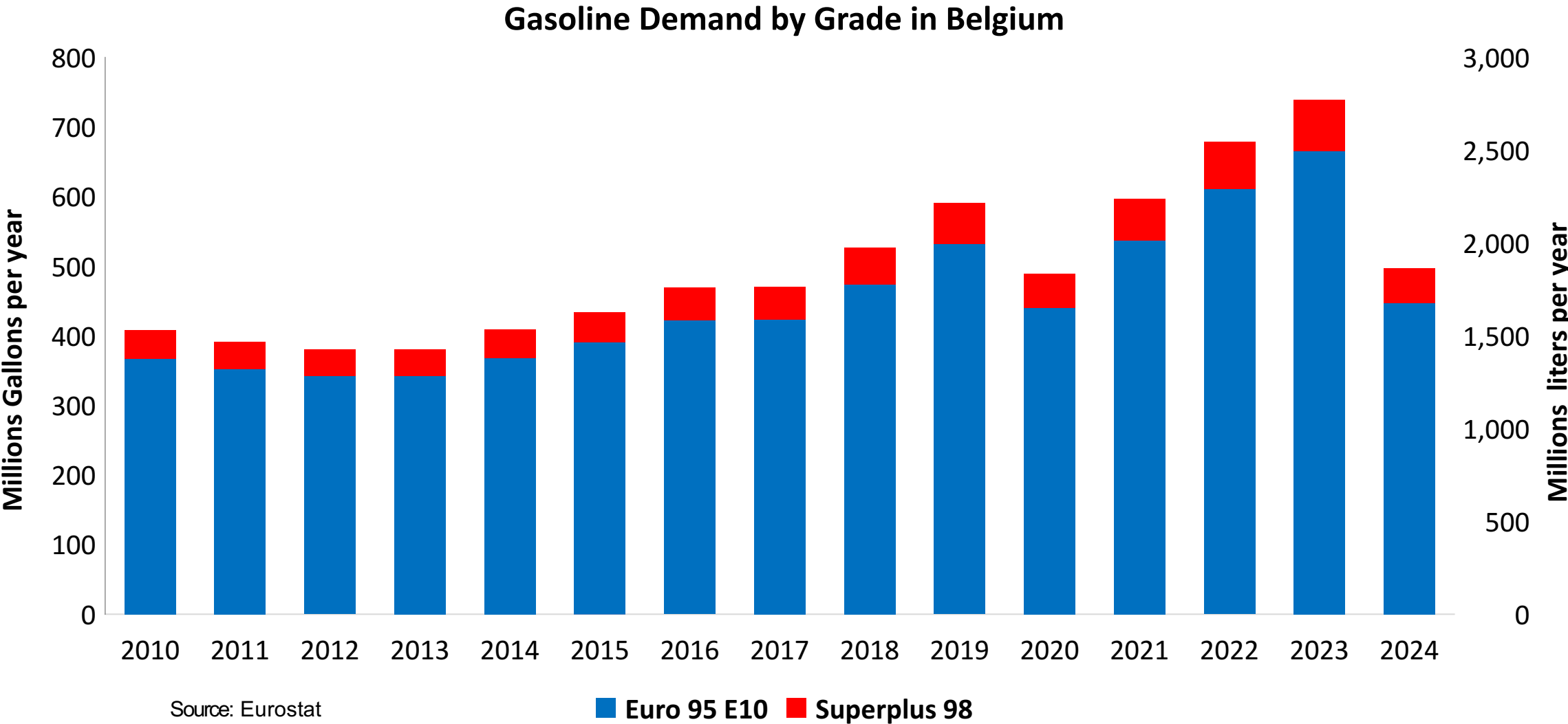


In 2024, gasoline consumption in Belgium was 1,865 million liters (493 million gallons) and growth rate was -3.4%. Gasoline production and net imports were 5,257 and -4,825 million liters (1,389 and -1,275 million gallons) correspondingly. Belgium complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Belgium was 239 million liters (63 million gallons) representing 12.8% of gasoline demand. Ethanol production and Net Imports were 554 and -315 million liters (63 and -83 million gallons) correspondingly. Ethanol sources are Sugar Cane 20% and Cereals 80%.

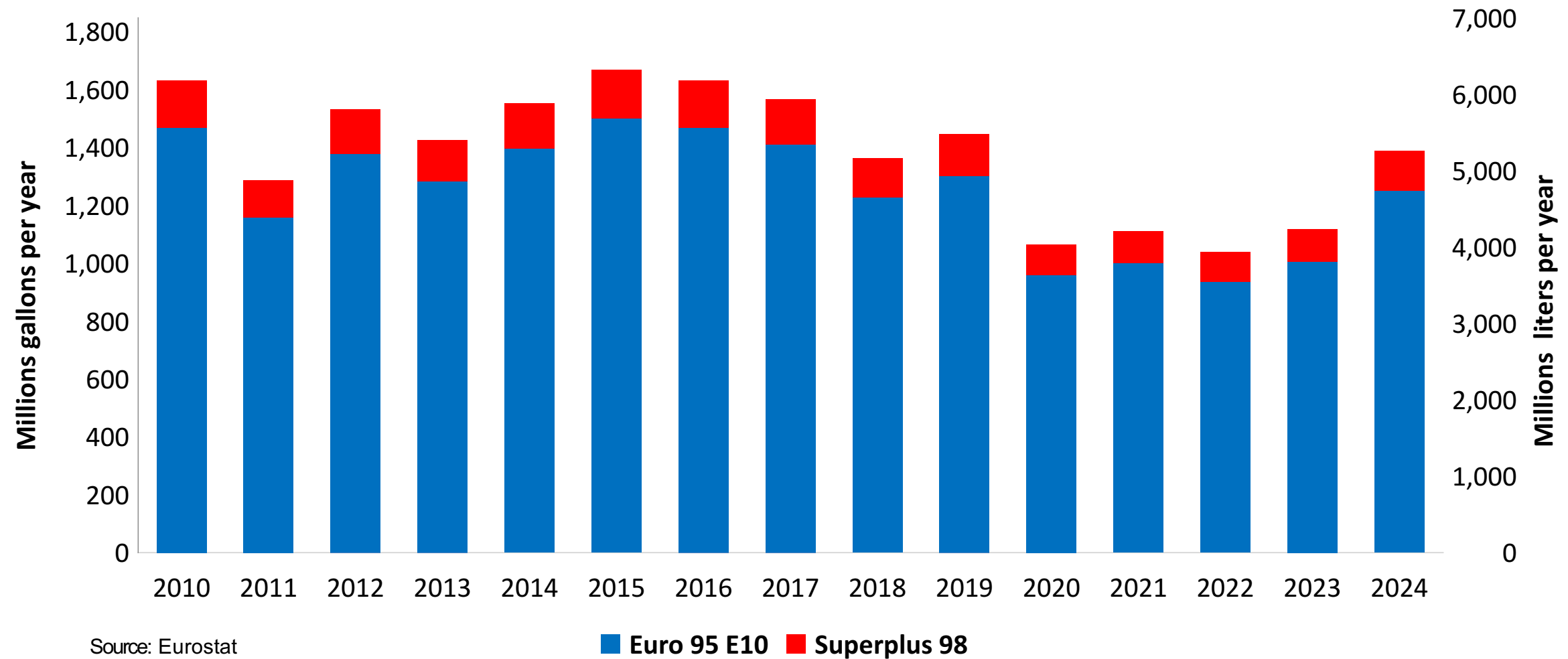
Source: Eurostat, European Commission

Gasoline Demand in Belgium



Gasoline Production in Belgium

Gasoline Production by Grade in Belgium



Source: Eurostat

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene ring. The structure features a cyclohexene ring with a methyl group, a vinyl group, and a substituent labeled 'X'.

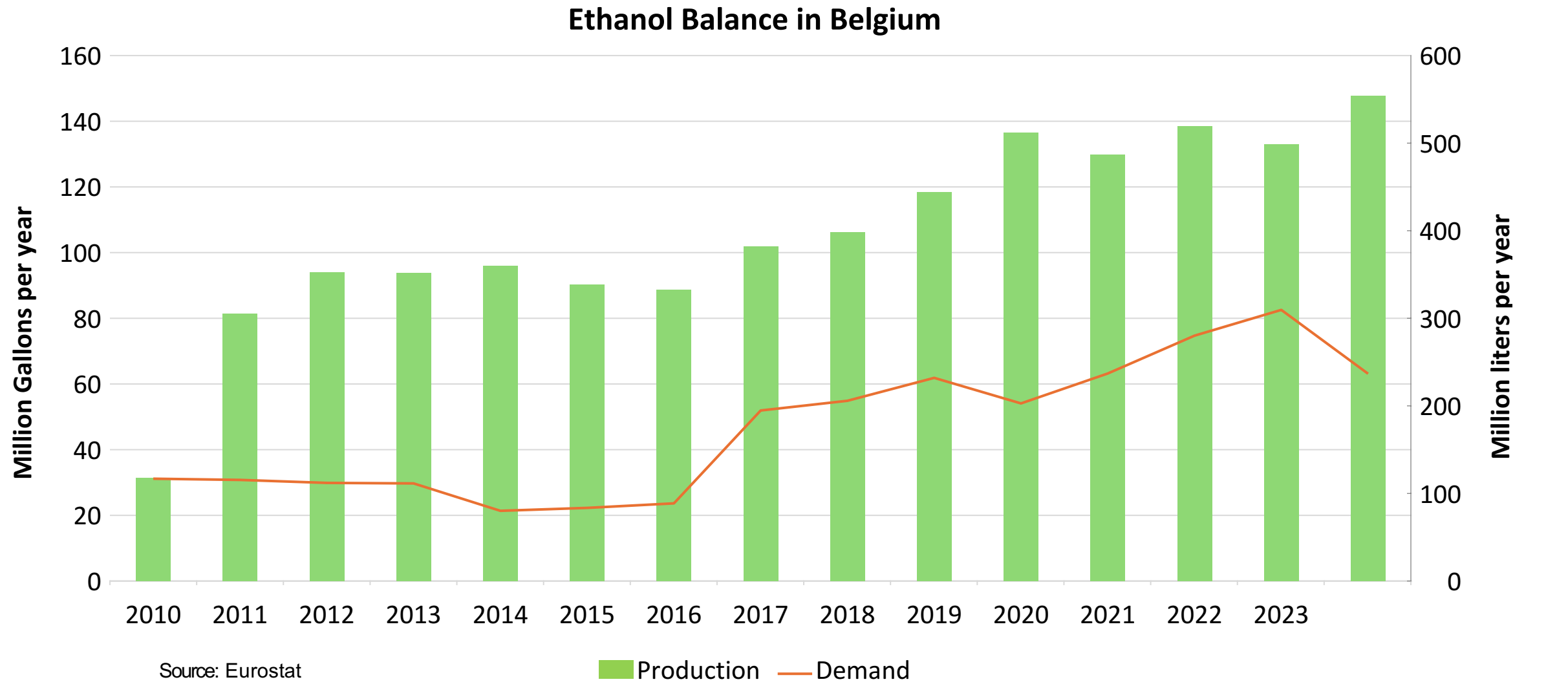
Year	Euro 95 E10 (Millions gallons)	Superplus 98 (Millions gallons)	Total (Millions gallons)	Total (Millions liters)
2010	325	35	360	1,380
2011	355	40	395	1,500
2012	315	35	350	1,320
2013	260	30	290	1,100
2014	275	30	305	1,160
2015	265	30	295	1,120
2016	250	28	278	1,060
2017	270	30	300	1,140
2018	165	20	185	700
2019	160	18	178	670
2020	130	15	145	550
2021	60	5	65	250
2022	125	15	140	530
2023	75	10	85	320
2024	215	25	240	910

Source: Eurostat

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexene. The structure features a six-membered ring with a double bond. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.



Ethanol Balance in Belgium



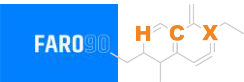
Gasoline Quality in Belgium

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.5							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	5.7							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

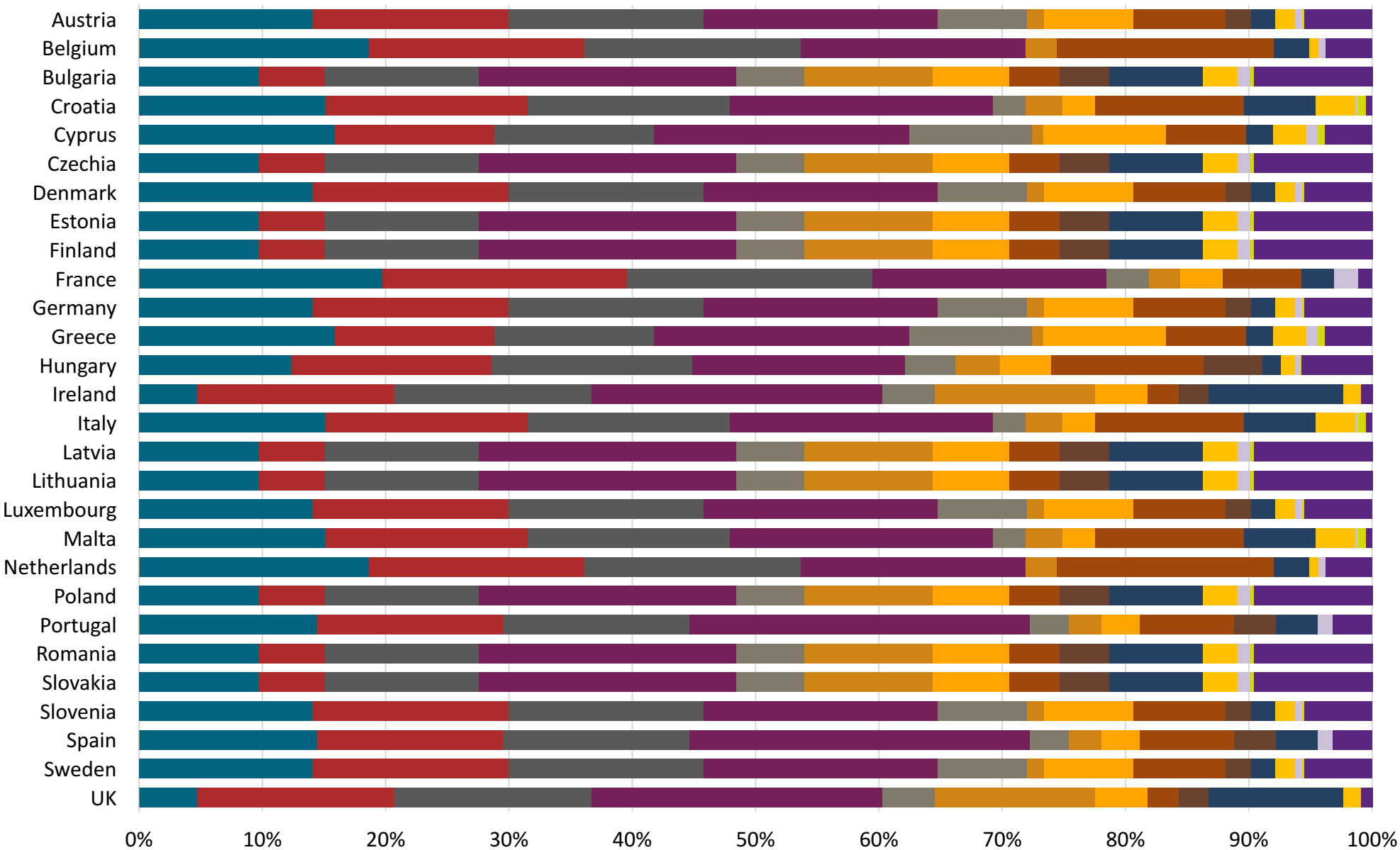
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



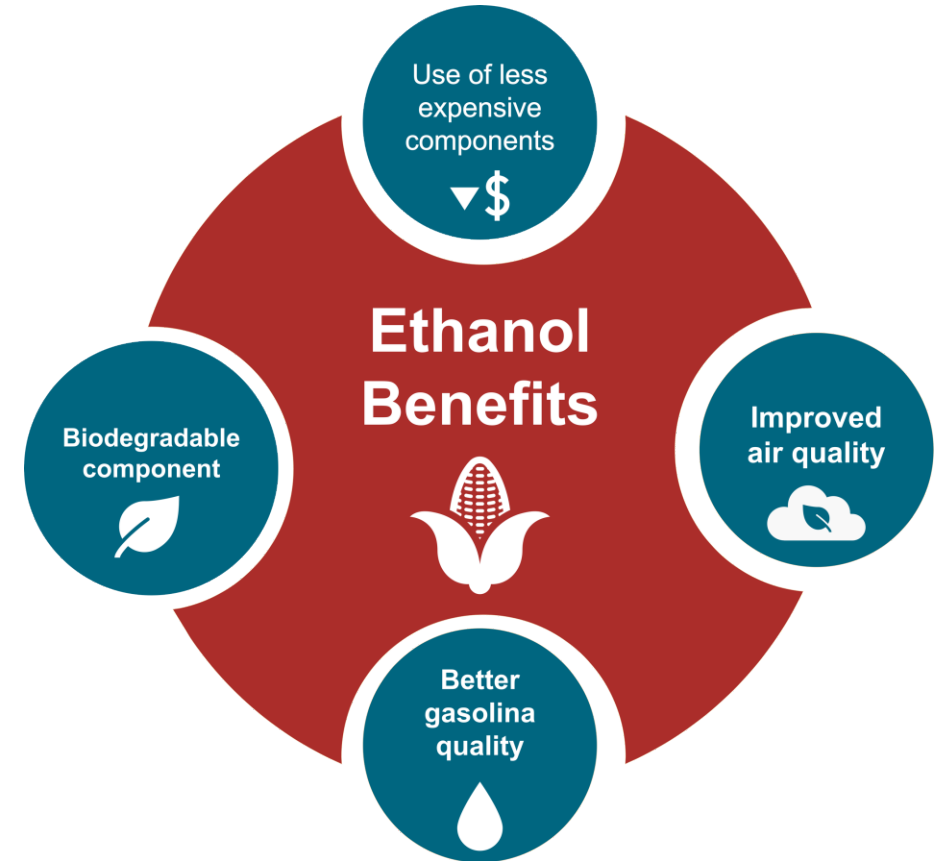
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

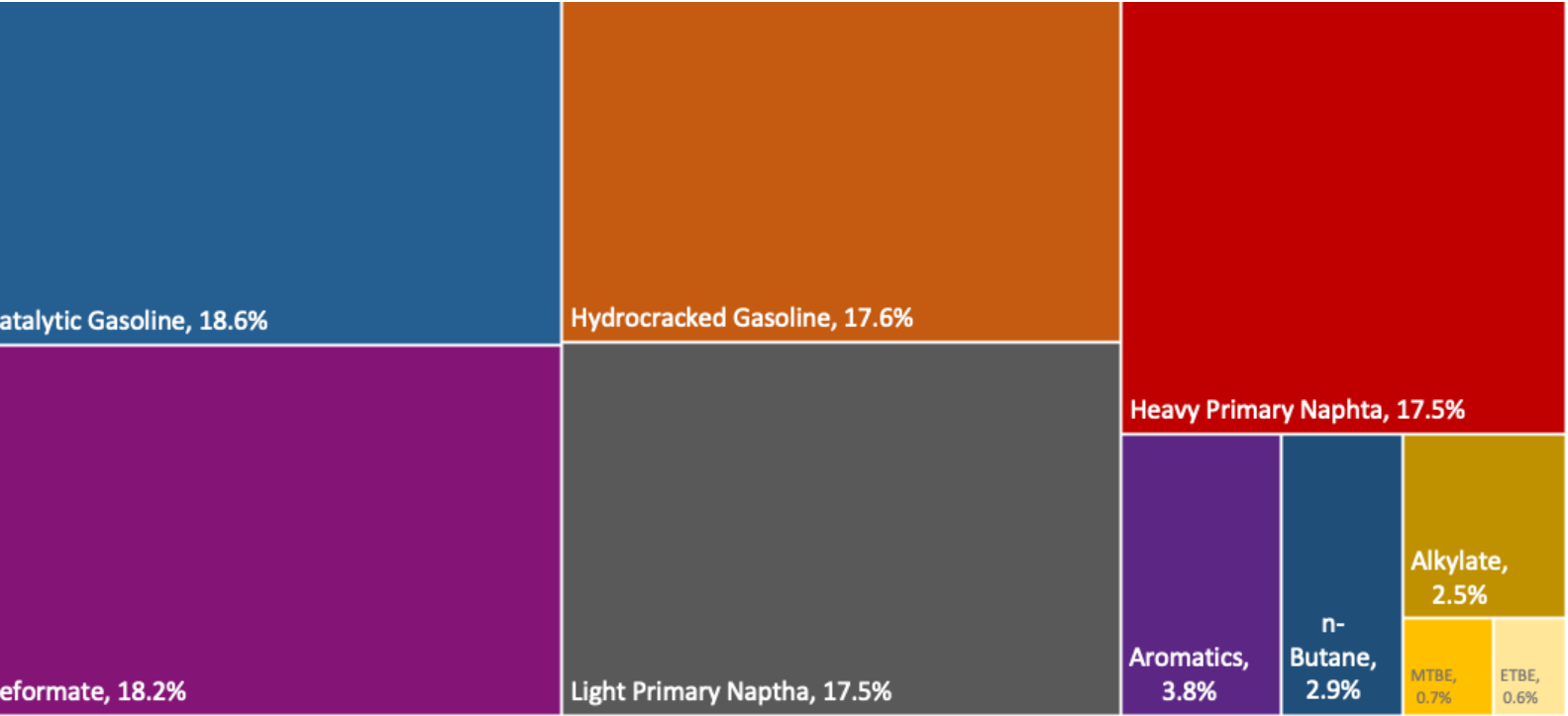
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

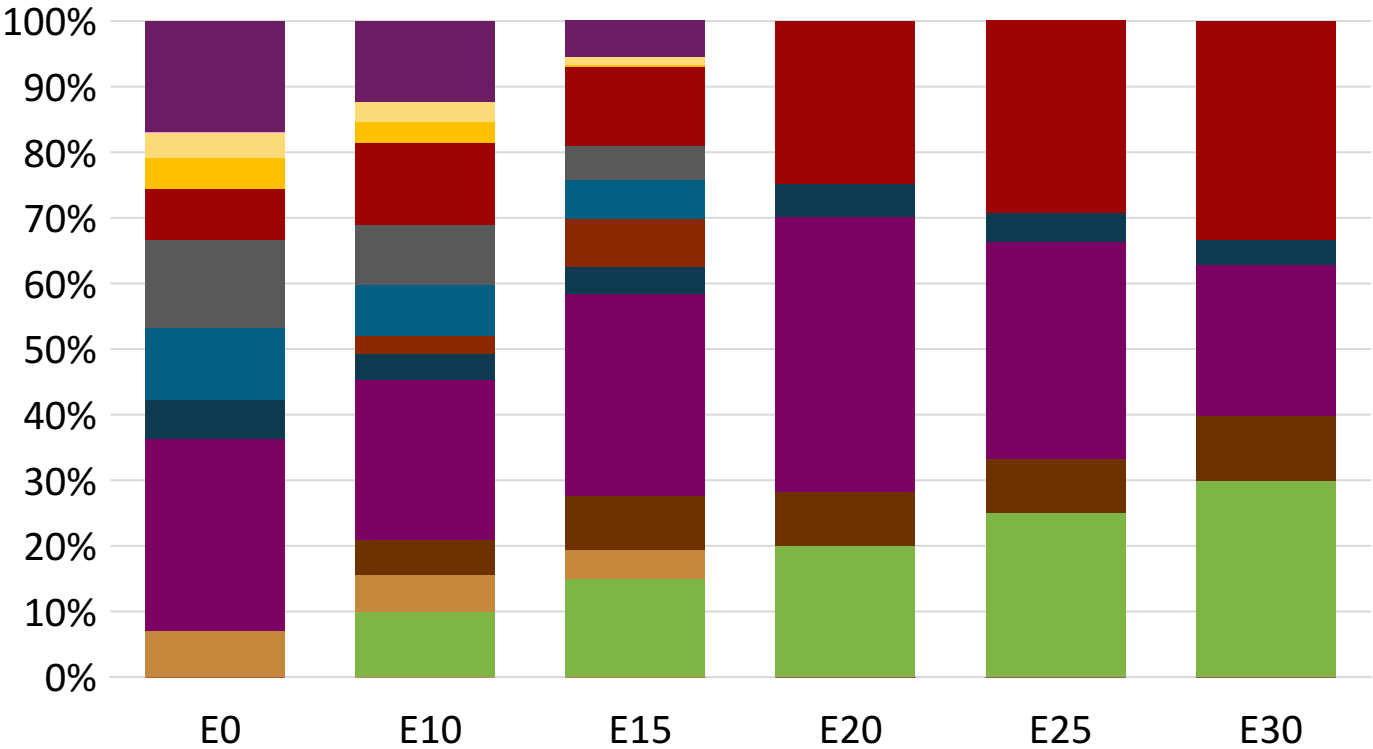
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Belgium Available Blending Components



Belgium – Gasoline 95 – Constant Octane

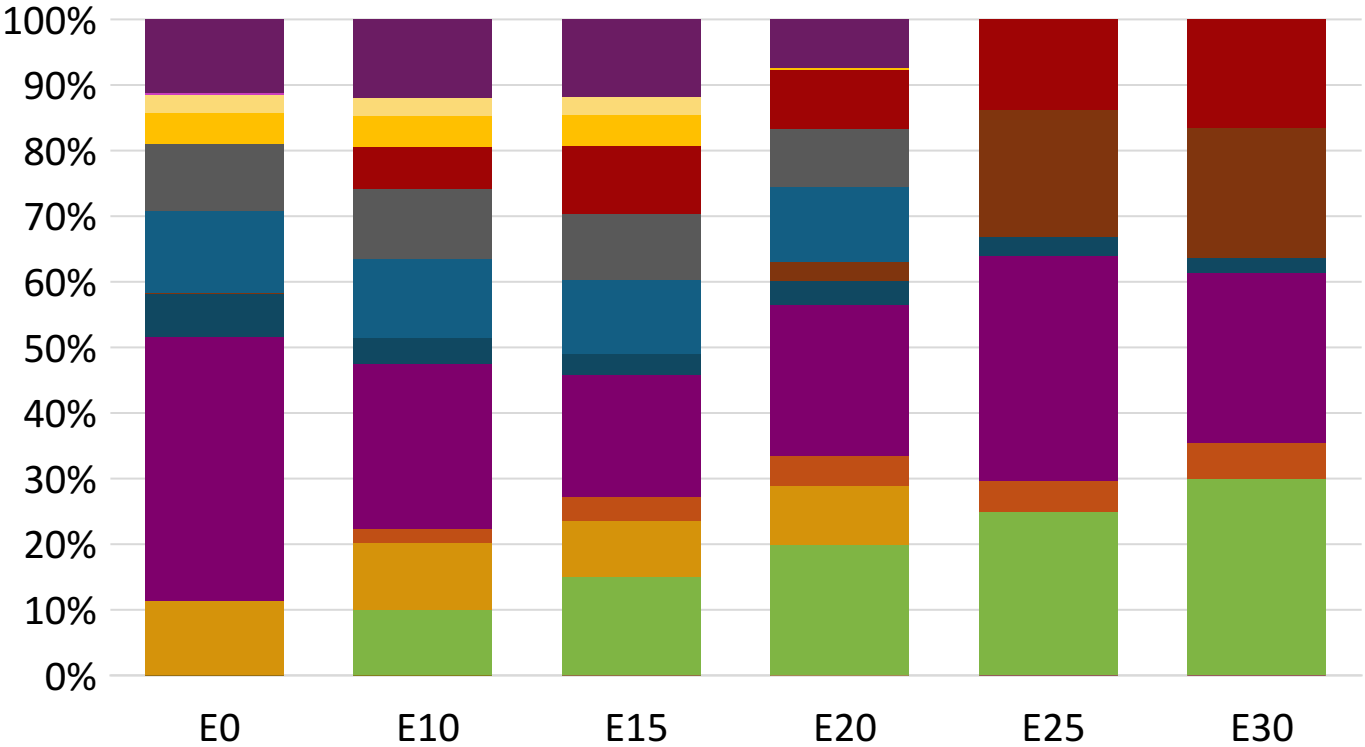
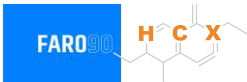


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US\$/gal)	\$ 2.281	\$ 2.147	\$ 2.072	\$ 1.951	\$ 1.875	\$ 1.780

Source: Faro90

Belgium - Gasoline 98 - Constant Octane

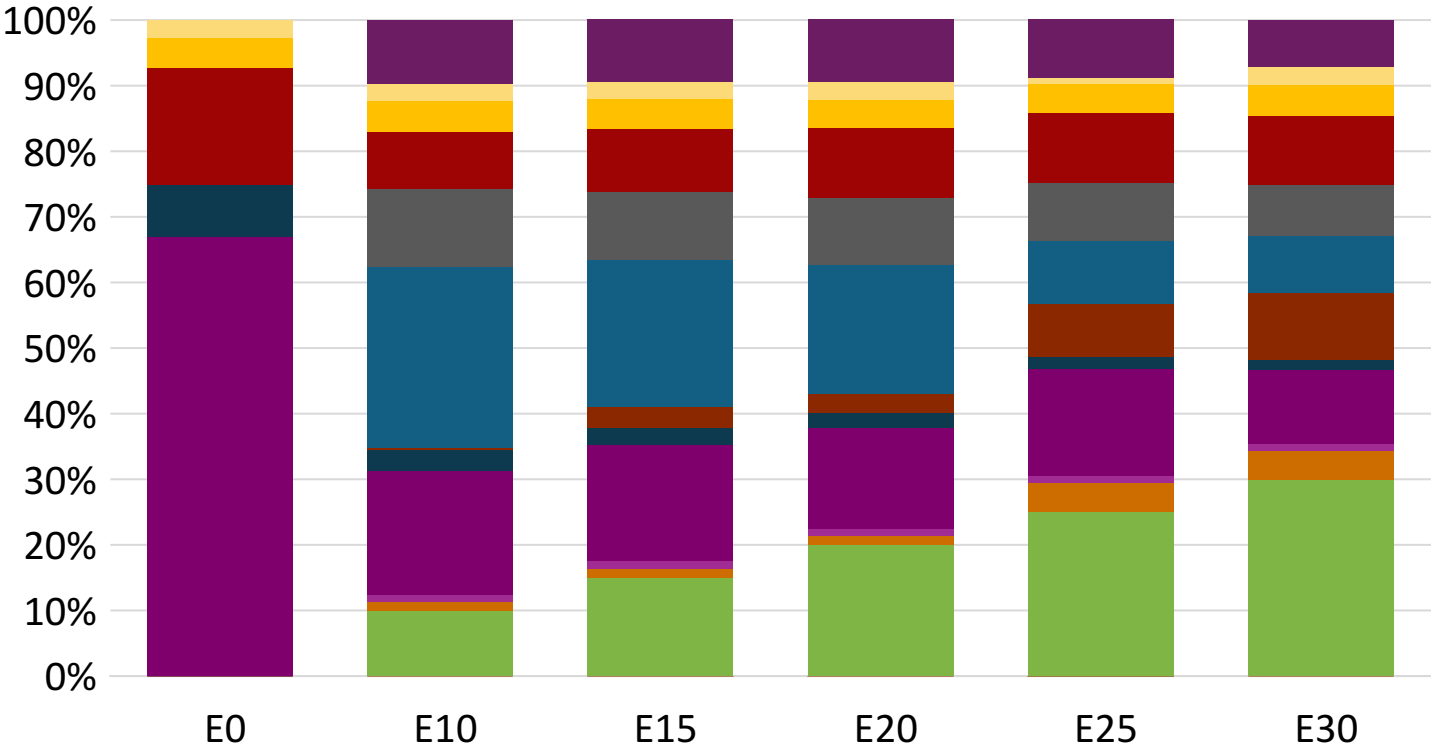


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US\$/gal)	\$ 2.409	\$ 2.290	\$ 2.216	\$ 2.164	\$ 2.057	\$ 1.994

Source: Faro90

Belgium – Gasoline 95 – Increased Octane

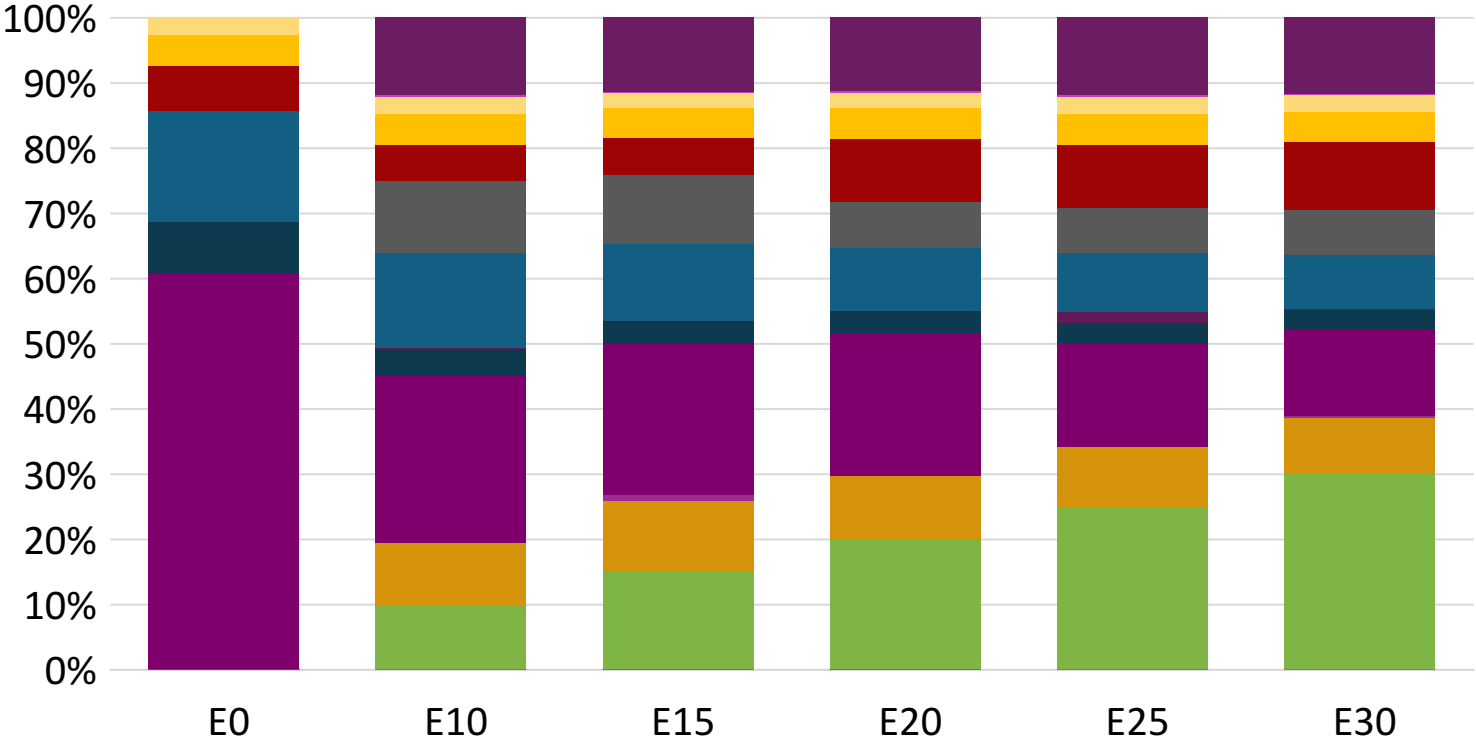


Average CIF 2024 Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US\$/gal)	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193

Source: Faro90

Belgium – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)	98.0	98.8	100.1	101.6	102.9	104.2
Price (US\$/gal)	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

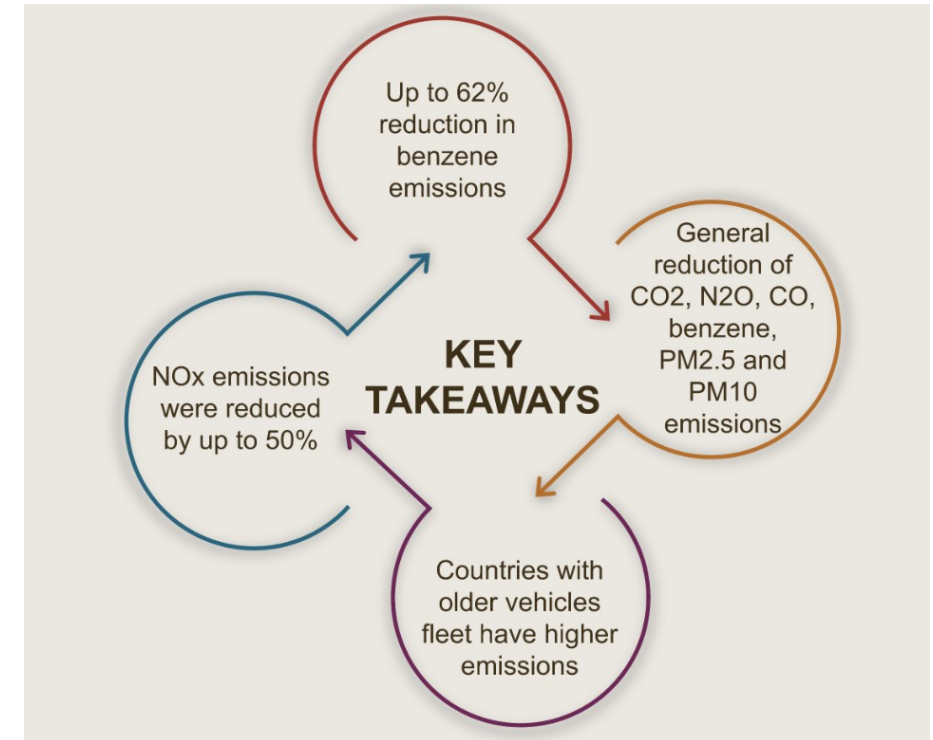
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

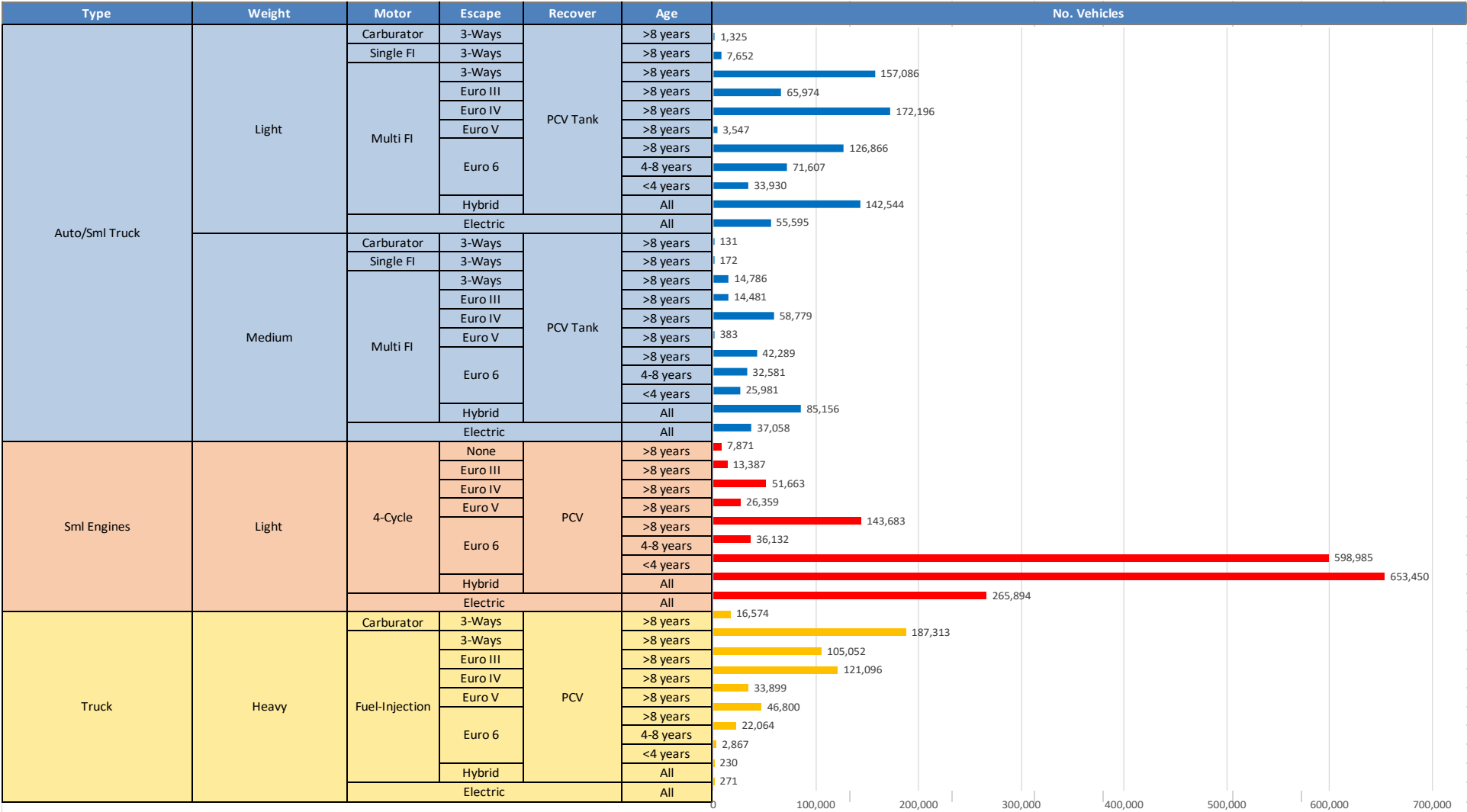
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



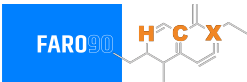
*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Belgium

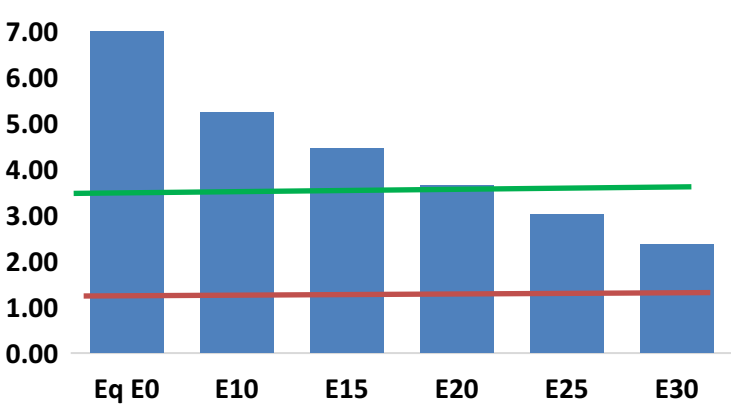


Belgium – Vehicular Emissions

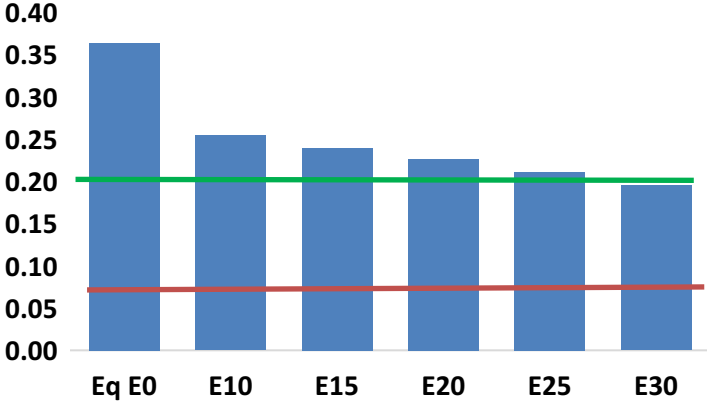
TIER III BIN 125 United States
Euro 6 Standard



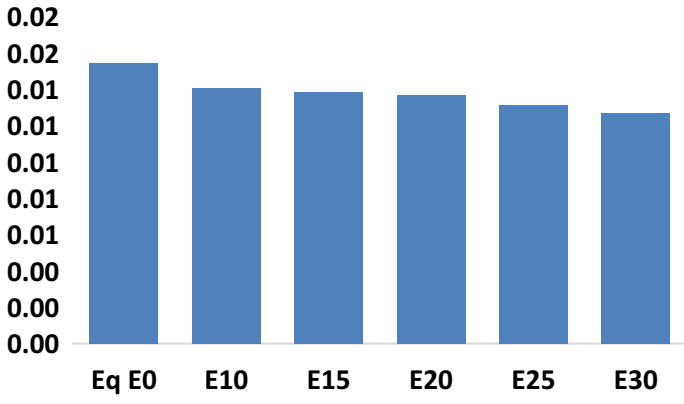
Carbon Monoxide (CO) g/km



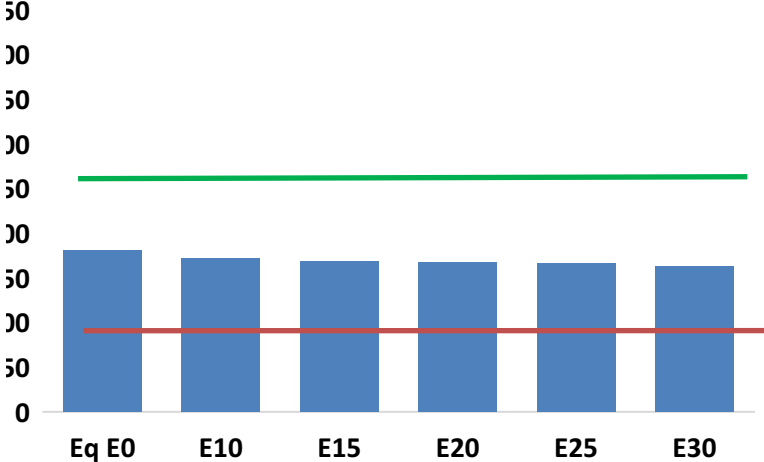
Nitrogen Oxides (NOx) g/km



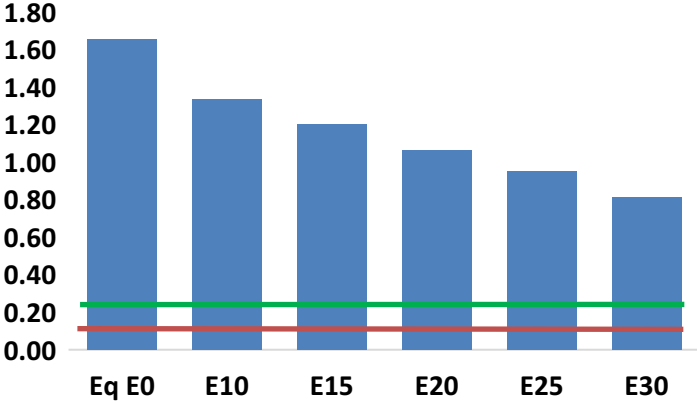
Aromatics (BTX) g/km



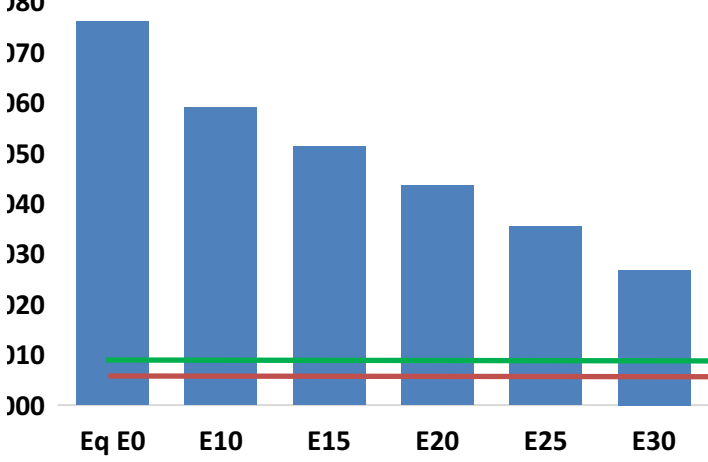
Carbon Dioxide (CO2) g/km



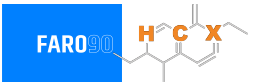
Volatile Organic Compounds (VOC) g/km



Particulate Matter (PM2.5) g/km

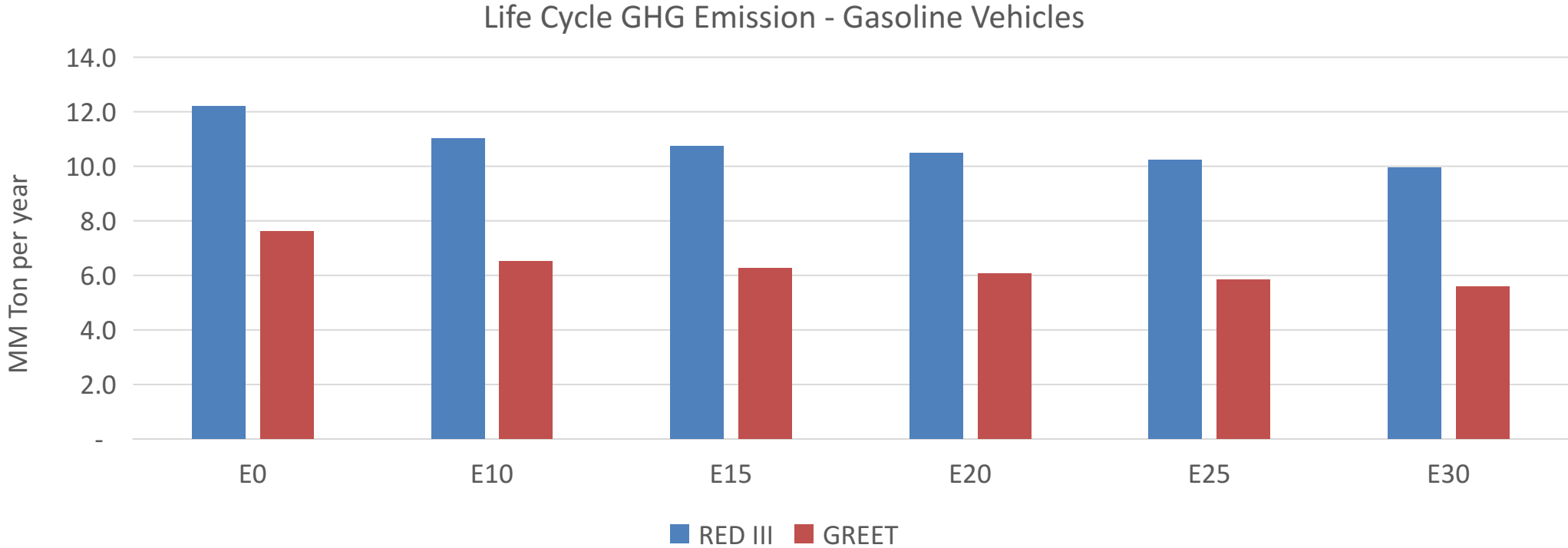


Belgium – Vehicular Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	424,560	371,491	318,422	270,948	221,338	182,637	143,912	-25%	-48%
CO2	10,967,384	10,685,366	10,419,015	10,209,538	10,105,881	10,047,600	9,862,394	-5%	-8%
VOC	99,964	90,398	80,831	72,624	64,196	57,665	49,129	-19%	-36%
NOx	21,975	16,481	15,382	14,498	13,673	12,763	11,801	-30%	-38%
SOx	124	115	105	97	90	81	73	-15%	-28%
NH3	4,858	4,810	4,762	4,785	4,839	4,877	4,908	-2%	0%
N2O	207	205	205	206	210	212	215	-1%	2%
CH4	19,986	19,986	19,986	20,285	20,725	21,105	21,465	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	403	375	347	324	302	285	261	-14%	-25%
Acetaldehyde	1,575	1,808	2,653	4,040	5,504	6,290	7,432	68%	249%
Formaldehyde	6,299	6,124	7,138	8,382	8,781	9,715	10,575	13%	39%
Benzene	934	897	850	837	830	793	768	-9%	-11%
PM2.5	706	627	547	475	404	328	248	-22%	-43%
PM10	3,897	3,461	3,024	2,626	2,230	1,809	1,371	-22%	-43%

Belgium – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	105.6
Target @ 2035 (MMT/year)	59.4
Delta	46.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.3%	17.7%	20.5%	23.3%	26.8%
RED II/III	0.0%	9.5%	12.0%	14.0%	15.9%	18.3%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.4%	2.9%	3.4%	3.8%	4.4%
RED II/III	0.0%	2.5%	3.2%	3.7%	4.2%	4.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90